



A Molecular String Representations

SMILES

CNC(C)Cc1ccc2c...

C N C (C) C c 1 c c c 2 c ...
 ↓ ↓ ↓ ↓
 X X X X

SELFIES

[C] [N] [C] [Branch1] [C] [C] [C] [C] [=C] [C] [=C] [C] [=Branch1] [Ring2] ...

B Byte Pair Encoding (BPE)

Primitive SELFIES token

[C] [N] [C] [Branch1] [C] [C] [C] [C] [=C] ...



Frequency-based token merges

[C][N]][C]][Branch 1][C] [C][C]][C] ...

C Atom Pair Encoding (APE)

Primitive SELFIES token

[C] [N] [C] [Branch1] [C] [C] [C] [C] [=C] ...



APE-constrained merges

[C][N] [C][Branch1] [C][C] [C][C] [=C][C] ...